

STARAYA RUSSA, Russian Federation

WMO: 262750

Lat: 58.017N

Lon: 31.317E

Elev: 25

StdP: 101.03

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.6	-21.2	-26.9	0.3	-24.4	-23.3	0.5	-21.0	7.8	-0.7	7.1	-0.3	1.0	340	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.7	20.6	26.7	19.6	24.9	18.6	21.7	27.0	20.6	25.3	19.5	23.7	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.5	24.4	18.8	13.6	23.2	17.8	12.8	21.9	63.2	27.1	59.3	25.1	55.6	23.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.8	5.2	-27.3	31.0	4.4	1.9	-30.5	32.4	-33.1	33.6	-35.5	34.7	-38.7	36.1		
(5)			-27.5	23.0	4.4	1.6	-30.6	24.2	-33.2	25.1	-35.6	26.0	-38.8	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-6.1	-6.2	-1.2	6.3	12.2	15.9	18.7	16.8	11.8	5.7	0.2	-3.5	(6)
(7)		DBStd	10.03	7.06	7.30	4.88	4.36	4.35	3.58	3.21	3.10	3.60	4.11	5.06	6.13	(7)
(8)		HDD10.0	2337	500	453	348	128	27	2	0	1	23	142	295	418	(8)
(9)		HDD18.3	4609	758	687	607	362	194	90	34	67	198	392	545	676	(9)
(10)		CDD10.0	859	0	0	0	16	96	179	271	212	76	9	0	0	(10)
(11)		CDD18.3	89	0	0	0	0	4	17	46	21	1	0	0	0	(11)
(12)		CDH23.3	761	0	0	0	2	53	149	377	172	8	0	0	0	(12)
(13)		CDH26.7	168	0	0	0	0	9	31	93	34	0	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.7	2.6	2.7	2.5	2.3	2.1	1.8	1.9	2.0	2.4	2.8	3.0	(14)
(15)	Precipitation	PrecAvg	652	46	36	32	38	60	72	76	77	56	61	56	44	(15)
(16)		PrecMax	909	74	61	64	91	154	153	196	176	104	138	86	66	(16)
(17)		PrecMin	460	13	6	9	6	9	17	6	2	15	14	1	19	(17)
(18)		PrecStd	112	16	14	16	20	29	35	46	45	23	29	22	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.2	13.5	23.2	27.8	29.8	31.6	30.0	24.6	17.5	10.4	8.5	(19)
(20)			MCWB	4.4	3.2	7.3	13.3	18.9	20.5	22.5	21.9	18.4	12.9	9.0	7.2	(20)
(21)		2%	DB	3.5	3.9	9.6	19.6	24.5	27.3	29.2	27.2	21.5	14.5	8.8	6.4	(21)
(22)			MCWB	2.6	2.5	4.9	11.4	16.3	19.6	21.2	19.7	16.2	11.7	7.4	5.3	(22)
(23)		5%	DB	2.3	2.8	6.6	16.7	22.2	24.9	27.2	25.2	19.4	12.9	7.5	4.7	(23)
(24)			MCWB	1.5	1.6	3.6	9.9	14.8	18.1	20.4	18.8	15.1	10.9	6.4	3.9	(24)
(25)		10%	DB	1.5	1.9	4.7	14.0	20.0	23.0	25.3	23.2	17.6	11.6	6.2	3.3	(25)
(26)			MCWB	0.8	1.0	2.5	8.7	13.6	16.9	19.4	18.0	14.2	10.0	5.3	2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	3.8	7.8	14.3	20.5	22.0	23.9	22.6	18.9	13.7	9.3	7.3	(27)
(28)			MCD B	5.2	4.7	12.9	21.2	26.7	28.2	29.9	28.3	23.5	15.9	10.0	8.2	(28)
(29)		2%	WB	2.7	2.6	5.4	12.2	17.7	20.4	22.1	21.0	17.1	12.2	7.7	5.3	(29)
(30)			MCD B	3.3	3.6	8.7	17.9	22.4	25.4	27.6	25.5	20.3	14.0	8.5	6.1	(30)
(31)		5%	WB	1.6	1.8	4.0	10.7	16.0	19.1	21.0	19.7	15.8	11.3	6.5	4.0	(31)
(32)			MCD B	2.2	2.7	6.1	15.7	20.3	23.4	25.9	23.5	18.5	12.8	7.3	4.7	(32)
(33)		10%	WB	0.9	1.1	2.8	9.2	14.6	17.7	20.0	18.5	14.6	10.1	5.4	2.5	(33)
(34)			MCD B	1.4	1.9	4.4	13.6	18.9	21.8	24.4	22.4	17.1	11.6	6.2	3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.9	6.2	7.3	9.5	10.4	9.5	9.7	8.4	5.7	4.0	4.1	(35)	
(36)			MCDBR	3.4	4.3	9.6	14.2	13.9	12.4	12.3	12.9	11.3	7.5	4.6	3.7	(36)
(37)			MCWBR	3.2	3.4	6.2	7.4	7.0	6.0	5.8	6.4	6.6	5.3	4.2	3.5	(37)
(38)		5% WB	MCDBR	3.3	3.6	8.3	12.3	11.5	10.8	11.1	11.0	9.9	6.6	4.4	3.6	(38)
(39)			MCWBR	3.2	3.1	5.7	6.8	6.7	5.5	5.8	6.2	6.3	5.0	4.2	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.322	0.343	0.386	0.376	0.372	0.398	0.399	0.365	0.343	0.320	0.277	(40)	
(41)		taud	2.117	2.110	2.181	2.268	2.379	2.413	2.356	2.344	2.414	2.372	2.215	2.115	(41)	
(42)		Ebn at Noon	536	685	791	820	858	866	831	800	773	675	493	423	(42)	
(43)		Edh at Noon	53	85	104	114	109	107	111	105	84	66	49	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.38	1.09	2.49	3.82	5.22	5.42	5.24	4.16	2.61	1.12	0.36	0.21	(44)	
(45)		RadStd	0.08	0.20	0.34	0.48	0.49	0.51	0.58	0.36	0.29	0.20	0.06	0.05	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (8 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-158	-181	N/A	N/A	

Nomenclature: See separate page